Today

- Operators and wavefunctions in momentum representation
- General uncertainty principle
- The simple harmonic oscillator and number state vector space

3.4.1 Operators and wavefunction in position representation

Now suppose the state vector of a system in Hilbert space is given by $|\psi(t)\rangle$. The position and the momentum operators in position space are given by

$$\hat{\vec{r}} = \vec{r}, \qquad \hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{3.4.27}$$

A physical observable in position representation is a function of position and momentum operators in position representation,

$$\hat{O}(\hat{\vec{r}}, \hat{\vec{p}}) = \hat{O}(\vec{r}, -i\hbar\vec{\nabla}). \tag{3.4.28}$$

The expectation value $\hat{O}$ is a measurable quantity that should be independent of the representation. In state $\psi(t)$,

$$\langle\hat{O}\rangle = \langle\psi(t)|\hat{O}|\psi(t)\rangle. \tag{3.4.29}$$

To write the expectation value in position representation, we insert $\int d^2\vec{r}\,|\vec{r}\rangle\langle\vec{r}| = 1$, and we have

$$\langle\hat{O}\rangle = \int d^2\vec{r}\langle\psi(t)|\vec{r}\rangle\langle\vec{r}|\hat{O}|\psi(t)\rangle. \tag{3.4.30}$$

Thus,

$$\langle\psi(t)|\vec{r}\rangle = \langle\vec{r}|\psi(t)\rangle^* = \psi(\vec{r}, t)^* \tag{3.4.31}$$

and $\langle\vec{r}|\hat{O}|\psi(t)\rangle$ is just $\hat{O}|\psi(t)\rangle$ in position representation; i.e.,

$$\langle\vec{r}|\hat{O}|\psi(t)\rangle = \hat{O}(\vec{r}, -i\hbar\vec{\nabla})\psi(\vec{r}, t) \tag{3.4.32}$$

so that

$$\langle\hat{O}\rangle = \int d^3\vec{r}\psi(\vec{r}, t)^*\hat{O}(\vec{r}, -i\hbar\vec{\nabla})\psi(\vec{r}, t). \tag{3.4.33}$$

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In other words, given a state vector $|\psi(t)\rangle$ in Hilbert space, the wavefunction in position representation is

$$\psi(\vec{r}, t) = \langle \vec{r}|\psi(t)\rangle. \tag{3.4.34}$$

Question 3.4.3: Considering the unit vectors in three dimensions, what does $\hat{x}(\hat{x}\cdot) + \hat{y}(\hat{y}\cdot) + \hat{z}(\hat{z}\cdot)$ represent?

Question 3.4.4: Write down and discuss the wavefunctions of definite position and definite momentum in both position and momentum representation.

Question 3.4.5: What does the negative sign in the exponent in $\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar}px}$ indicate in contrast to the positive sign in contrast to $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar}px}$.

3.4.2 Operators and wavefunction in momentum representation

First, the momentum operator in momentum space is $\hat{\vec{p}} = \vec{p}$. To determine the position operator in momentum space, we seek an operator $\hat{\vec{r}}$ which produces a definite position when applied on a wavefunction of definite position in momentum space; i.e.,

$$\hat{\vec{r}} \frac{1}{(2\pi\hbar)^{3/2}} e^{-\frac{i}{\hbar}\vec{p}\cdot\vec{r}} = \vec{r} \frac{1}{(2\pi\hbar)^{3/2}} e^{-\frac{i}{\hbar}\vec{p}\cdot\vec{r}}, \tag{3.4.35}$$

and, therefore, the position operator in momentum space is given by

$$\hat{\vec{r}} = i\hbar\vec{\nabla}_{\vec{p}}, \tag{3.4.36}$$

where $\vec{\nabla}_{\vec{p}}$ is the gradient operator with respect to momentum. In particular, in one dimension,

$$\hat{x} = i\hbar\frac{\partial}{\partial p}, \tag{3.4.37}$$

In $\hat{\vec{p}}|\vec{p}\rangle = \vec{p}|\vec{p}\rangle$, $\hat{\vec{p}}$ is an operator, $|\vec{p}\rangle$ is a state vector, and $\vec{p}$ is an eigenvalue. In momentum representation, the eigenvalue $\vec{p}$ has real components by definition, since it represents the momentum of a particle. In other words, $\vec{p}^{\,*} = \vec{p}$.

Exercise 3.4.3: Show that the position operator in momentum representation is given by $\hat{x} = i\hbar\frac{\partial}{\partial p}$ directly using matrix elements.

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Solution:

$$\begin{aligned}
\langle\psi|\hat{x}|\psi\rangle &= \int dpdp'\langle\psi|p\rangle\langle p|\hat{x}|p'\rangle\langle p'|\psi\rangle \\
&= \int dpdp'dx\langle\psi|p\rangle\langle p|x\rangle\langle x|\hat{x}|p'\rangle\langle p'|\psi\rangle \\
&= \int dpdp'dx\, x\langle\psi|p\rangle\langle p|x\rangle\langle x|p'\rangle\langle p'|\psi\rangle \\
&= \int dpdp'dx\, x\langle\psi|p\rangle\frac{1}{\sqrt{2\pi\hbar}}e^{-\frac{i}{\hbar}px}\frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}p'x}\langle p'|\psi\rangle \\
&= \int dpdp'\langle\psi|p\rangle\left(\frac{1}{2\pi\hbar}\int dx\, xe^{-\frac{i}{\hbar}(p-p')x}\right)\langle p'|\psi\rangle \\
&= \int dpdp'\langle\psi|p\rangle\left(i\hbar\frac{\partial}{\partial p}\delta(p-p')\right)\langle p'|\psi\rangle. \qquad (3.4.38)
\end{aligned}$$

Comparing the first and the last lines, we have

$$\langle p|\hat{x}|p'\rangle = i\hbar\frac{\partial}{\partial p}\delta(p-p'). \qquad (3.4.39)$$

Thus, the position operator in momentum representation is given by $\hat{x} = i\hbar\frac{\partial}{\partial p}$.

Exercise 3.4.4: Show that the position operator in momentum space is Hermitian.

Solution: The solution is simply a repeat of Exercise 3.4.1 with $p \leftrightarrow x$ and $i \leftrightarrow -i$.

An observable in momentum representation is a function of position and momentum operators in momentum representation,

$$\hat{O}(\hat{\vec{r}},\hat{\vec{p}}) = \hat{O}(i\hbar\vec{\nabla}_{\vec{p}},\vec{p}). \qquad (3.4.40)$$

Because the expectation value $\hat{O}$ is independent of the representation, in state $|\psi(t)\rangle$,

$$\begin{aligned}
\langle\hat{O}\rangle &= \langle\psi(t)|\hat{O}|\psi(t)\rangle = \int d^2\vec{p}d^2\vec{p}'\langle\psi(t)|\vec{p}\rangle\langle\vec{p}|\hat{O}|\vec{p}'\rangle\langle\vec{p}'|\psi(t)\rangle \\
&= \int d^2\vec{p}d^2\vec{p}'\langle\psi(t)|\vec{p}\rangle\hat{O}(i\hbar\vec{\nabla}_{\vec{p}},\vec{p})\langle\vec{p}|\vec{p}'\rangle\langle\vec{p}'|\psi(t)\rangle \\
&= \int d^2\vec{p}d^2\vec{p}'\psi(\vec{p},t)^*\hat{O}(i\hbar\vec{\nabla}_{\vec{p}},\vec{p})\delta^3(\vec{p}-\vec{p}')\psi(\vec{p}',t)\rangle \\
&= \int d^3\vec{p}\psi(\vec{p},t)^*\hat{O}(i\hbar\vec{\nabla}_{\vec{p}},\vec{p})\psi(\vec{p},t). \qquad (3.4.41)
\end{aligned}$$

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Given a state vector $|\psi(t)\rangle$ in Hilbert space, the wavefunction in momentum representation is

$$\psi(\vec{p},t) = \langle\vec{p}|\psi(t)\rangle. \tag{3.4.42}$$

Because both $\hat{\vec{r}}$ and $\hat{\vec{p}}$ are Hermitian and time is not an operator, a physical observable $\hat{O}(\hat{\vec{r}},\hat{\vec{p}},t)$ that is a function of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ is Hermitian too. In particular, the Hamiltonian operator for energy $\hat{H}(\hat{\vec{r}},\hat{\vec{p}},t)$ is Hermitian.

Exercise 3.4.5: Show that the Schrödinger $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$, where $\hat{H} = \frac{\hat{\vec{p}}^2}{2m} + V(\hat{\vec{r}},t)$ becomes

1. $$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t) \tag{3.4.43}$$
   in position representation, and

2. $$i\hbar\frac{\partial\psi(\vec{p},t)}{\partial t} = -\frac{\vec{p}^2}{2m}\psi(\vec{p},t) + V(i\hbar\vec{\nabla}_{\vec{p}},t)\psi(\vec{p},t) \tag{3.4.44}$$
   in momentum representation.

Solution:

1. Inserting $1 = \int d^3\vec{r}|\vec{r}\rangle\langle\vec{r}|$ in $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$ we have $i\hbar\frac{\partial}{\partial t}\int d^3\vec{r}|\vec{r}\rangle\langle\vec{r}|\psi(t)\rangle = \int d^3\vec{r}d^3\vec{r}'|\vec{r}\rangle\langle\vec{r}|(\frac{\hat{\vec{p}}^2}{2m} + V(\hat{\vec{r}},t))|\vec{r}'\rangle\langle\vec{r}'|\psi(t)\rangle = \int d^3\vec{r}d^3\vec{r}'|\vec{r}'\rangle(-\frac{\hbar^2\vec{\nabla}^2}{2m} + V(\vec{r},t))\delta^3(\vec{r}-\vec{r}')\langle\vec{r}|\psi(t)\rangle = \int d^3\vec{r}|\vec{r}\rangle(-\frac{\hbar^2\vec{\nabla}^2}{2m} + V(\vec{r},t))\langle\vec{r}|\psi(t)\rangle$. Therefore, $\int d^3\vec{r}|\vec{r}\rangle\Big(i\hbar\frac{\partial}{\partial t}\langle\vec{r}|\psi(t)\rangle - (-\frac{\hbar^2\vec{\nabla}^2}{2m} + V(\vec{r},t))\langle\vec{r}|\psi(t)\rangle\Big) = 0$. Therefore, with $\langle\vec{r}|\psi(t)\rangle = \psi(\vec{r},t)$

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = -\frac{\hbar^2\vec{\nabla}^2}{2m}\psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t). \tag{3.4.45}$$

2. Inserting $1 = \int d^3\vec{p}|\vec{p}\rangle\langle\vec{p}|$ in $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$ we have $i\hbar\frac{\partial}{\partial t}\int d^3\vec{p}|\vec{p}\rangle\langle\vec{p}|\psi(t)\rangle = \int d^3\vec{p}d^3\vec{p}'|\vec{p}\rangle\langle\vec{p}|(\frac{\hat{\vec{p}}^2}{2m} + V(\hat{\vec{r}},t))|\vec{p}'\rangle\langle\vec{p}'|\psi(t)\rangle = \int d^3\vec{p}d^3\vec{p}'|\vec{p}'\rangle(\frac{\vec{p}^2}{2m} + V(i\hbar\vec{\nabla}_{\vec{p}},t))\delta^3(\vec{p}-\vec{p}')\langle\vec{p}|\psi(t)\rangle = \int d^3\vec{p}|\vec{p}\rangle(\frac{\vec{p}^2}{2m} + V(i\hbar\vec{\nabla}_{\vec{p}},t))\langle\vec{p}|\psi(t)\rangle$. Therefore, $\int d^3\vec{p}|\vec{p}\rangle\Big(i\hbar\frac{\partial}{\partial t}\langle\vec{p}|\psi(t)\rangle - (\frac{\vec{p}^2}{2m} + V(i\hbar\vec{\nabla}_{\vec{p}},t))\langle\vec{p}|\psi(t)\rangle\Big) = 0$. Therefore, with $\langle\vec{p}|\psi(t)\rangle = \psi(\vec{p},t)$,

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{p},t) = \frac{\vec{p}^2}{2m}\psi(\vec{p},t) + V(i\hbar\vec{\nabla}_{\vec{p}},t)\psi(\vec{p},t). \tag{3.4.46}$$

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Question 3.4.6: The Schrödinger equation for the simple harmonic oscillator in one dimension in momentum representation is (A). $i\hbar\frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x,t)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x,t)$, (B). $i\hbar\frac{\partial}{\partial t}\psi(p,t) = -\frac{\hbar^2}{2m}\frac{d^2\psi(p,t)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(p,t)$, (C). $i\hbar\frac{\partial}{\partial t}\psi(p,t) = \frac{p^2}{2m}\psi(p,t) - \frac{1}{2}m\hbar^2\omega^2\frac{d^2\psi(p,t)}{dp^2}$.

Question 3.4.7: Write down the wavefunction of the simple harmonic oscillator in momentum space using your knowledge of the wavefunctions in position space.

Exercise 3.4.6:

In problem 2.10.1, we investigated the Gaussian wave packet $\psi(x,t)$ in position representation (or position space), given $\psi(x,0) = Ae^{-\frac{x^2}{2a^2}}e^{ik_0x}$ with $A = \frac{1}{\sqrt{\sqrt{\pi}\,a}}$ when normalized. Now do the calculations below in the momentum representation (or momentum space) starting with $\phi(k,0) = \sqrt{\frac{a}{\sqrt{\pi}}}e^{-\frac{1}{2}a^2(k-k0)^2}$ that we obtained in 2.10.1(b).

1. Determine $\langle\hat{p}\rangle$ and $\langle\hat{p}^2\rangle$ at time t.
2. Determine $\langle\hat{x}\rangle$ and $\langle\hat{x}^2\rangle$ at time t.
3. Check that the uncertainty principle is satisfied with the minimum uncertainty saturated at $t = 0$.

3.5 Commuting operators

Consider a physical observable represented by Hermitian operator $\hat{A}$ for a system. Suppose we have solved the eigenvalue problem

$$\hat{A}|\psi\rangle = a|\psi\rangle; \tag{3.5.1}$$

i.e., we have found the eigenstates $|\psi\rangle$ and the corresponding real eigenvalues a of $\hat{A}$. Assume that the states are nondegenerate, with one eigenstate corresponding to one eigenvalue.

Next, consider a second physical observable represented by Hermitian operator $\hat{B}$. Suppose $\hat{A}$ and $\hat{B}$ commute; i.e.,

$$[\hat{A},\hat{B}] = 0. \tag{3.5.2}$$

We then have

$$\hat{A}\left(\hat{B}|\psi\rangle\right)\big) = \hat{B}\left(\hat{A}|\psi\rangle\right) = \hat{B}\left(a|\psi\rangle\right) = a\left(\hat{B}|\psi\rangle\right). \tag{3.5.3}$$

Therefore, $\hat{B}|\psi\rangle$ is an eigenstate of $\hat{A}$ with the same eigenvalue a as eigenstate $|\psi\rangle$. Because the states are nondegenerate and there is only one state with eigenvalue a, $\hat{B}|\psi\rangle$ and $|\psi\rangle$ can only

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differ by a constant, and we write

$$\hat{B}|\psi\rangle = b|\psi\rangle, \tag{3.5.4}$$

where b is a constant. In other words, $|\psi\rangle$ is a simultaneous eigenstate of $\hat{A}$ and $\hat{B}$. Because $|\psi\rangle$ is a simultaneous eigenstate of $\hat{A}$ and $\hat{B}$ with respective eigenvalues a and b, we rewrite it as

$$|\psi\rangle = |ab\rangle, \tag{3.5.5}$$

and we have

$$\hat{A}|ab\rangle = a|ab\rangle, \qquad \hat{B}|ab\rangle = b|ab\rangle. \tag{3.5.6}$$

Question 3.5.1: Suppose the simultaneous eigenstates of two commuting operators $\hat{A}$ and $\hat{B}$ are $|a_i b_i\rangle$. If a measurement of $\hat{A}$ is done and a_8 is obtained, then a following measurement of $\hat{B}$ in the same system produces (A). the ground state value of a, (B). 8, (C). b_8, (D). any value of b_i is possible.

Question 3.5.2: Suppose $\hat{A}$ and $\hat{B}$ do not commute. A first measurement of $\hat{A}$ is done and a_8 is obtained. A following measurement of $\hat{B}$ in the same system is done. What is the result of the measurement of $\hat{B}$? (A). the ground state value of a, (B). a_8, (C). b_8, (D). any eigenvalue of $\hat{B}$ is possible.

Question 3.5.3: Compare and discuss how the above two questions compare to $\hat{H}^n\psi_1(x,t)$ being a stationary state while $\hat{x}\psi_1(x,t)$ is not as we investigated for the particle in the infinite square well.

Exercise 3.5.1:

1. Consider two Hermitian operators $\hat{A}$ and $\hat{B}$ with simultaneous eigenvectors such that $\hat{A}|ab\rangle = a|ab\rangle$ and $\hat{B}|ab\rangle = b|ab\rangle$.
   (a) Show that $[\hat{A}, \hat{B}] = 0$. (5 points)
   (b) Show that $[\hat{H}, \hat{P}] = 0$ for a particle in a symmetric potential, where $\hat{H}$ is the Hamiltonian operator and $\hat{P}$ is the parity operator. Considering $[\hat{H}, \hat{P}] = 0$ and $\hat{P}^2 = 1$, what can we say about the symmetry of the wavefunctions in position representation? (5 points)
2. Suppose $[\hat{A}, \hat{B}] = 0$ and the eigenvalue a of $\hat{A}$ is 2-fold degenerate, with two orthonormal states corresponding to the same eigenvalue, $\hat{A}|ab^{(1)}\rangle = a|ab^{(1)}\rangle$ and $\hat{A}|ab^{(2)}\rangle = a|ab^{(2)}\rangle$.

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(a) Suppose we write $\hat{B}|ab^{(1)}\rangle = b_{11}|ab^{(1)}\rangle + b_{12}|ab^{(2)}\rangle$ and $\hat{B}|ab^{(2)}\rangle = b_{21}|ab^{(1)}\rangle + b_{22}|ab^{(2)}\rangle$, where b_{ij} are constants. Construct two orthonormal states $|ab_1\rangle$ and $|ab_1\rangle$ that are linear combinations of $|ab^{(1)}\rangle$ and $|ab^{(2)}\rangle$ and simultaneous eigenstates of $\hat{A}$ and $\hat{B}$ with nondegenerate distinct eigenvalues b_1 and b_2 for $\hat{B}$, and determine b_1 and b_2 in terms of b_{ij}. (5 points) [This is an example of the Gram-Schmidt orthogonalization procedure.]

(b) Now suppose b_{ij} are such that after doing (2a) it turns out $b_1 = b_2 = b$ and the states are still degenerate, $|ab_1\rangle = |ab_2\rangle = |ab\rangle$. What conclusion could you make, and how would you proceed to construct states that are not completely degenerate? (5 points)

3.6 Non-commuting operators and the general uncertainty principle

Next let us consider two Hermitian operators $\hat{A}$ and $\hat{B}$ that may be commuting or not.

Theorem: For any two observables represented by $\hat{A}$ and $\hat{B}$, the uncertainty ΔA in the measurement of $\hat{A}$ and the uncertainty ΔB in the measurement of $\hat{B}$ in any state satisfy

$$(\Delta A)^2(\Delta B)^2 \geq \left(\frac{1}{2i}\langle[\hat{A},\hat{B}]\rangle\right)^2. \tag{3.6.1}$$

Proof: Define $\Psi = (\hat{A} - \langle\hat{A}\rangle)|\psi\rangle$ and $\Phi = (\hat{B} - \langle\hat{B}\rangle)|\psi\rangle$. We then have

$$\begin{aligned}
(\Delta A)^2(\Delta B)^2 &= \langle\Psi|\Psi\rangle\langle\Phi|\Phi\rangle \\
&\geq |\langle\Psi|\Phi\rangle|^2, \quad \text{(using the Schwartz inequality } |\vec{V}|^2|\vec{W}|^2 \geq (\vec{V}\cdot\vec{W})^2) \\
&= \left(\frac{1}{2}(\langle\Psi|\Phi\rangle + \langle\Phi|\Psi\rangle)\right)^2 + \left(\frac{1}{2i}(\langle\Psi|\Phi\rangle - \langle\Phi|\Psi\rangle)\right)^2 \\
&\geq \left(\frac{1}{2i}(\langle\Psi|\Phi\rangle - \langle\Phi|\Psi\rangle)\right)^2 \\
&= \left(\frac{1}{2i}(\langle\psi|(\hat{A} - \langle\hat{A}\rangle)(\hat{B} - \langle\hat{B}\rangle)|\psi\rangle - \langle\psi|(\hat{B} - \langle\hat{B}\rangle)(\hat{A} - \langle\hat{A}\rangle)\psi\rangle)\right)^2 \\
&= \left(\frac{1}{2i}(\langle[\hat{A},\hat{B}]\rangle)\right)^2
\end{aligned} \tag{3.6.2}$$

In particular, for $\hat{A} = \hat{x}$ and $\hat{B} = \hat{p}$, we have $[\hat{A},\hat{B}] = i\hbar$ and $\Delta x\Delta p \geq \frac{\hbar}{2}$.

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Therefore, if the operators representing physical observables do not commute, they cannot be simultaneously measured to arbitrary precession; when one is more precisely measured the other is less precisely measured. Consequently, we cannot construct simultaneous eigenstates (which correspond to definite eigenvalues) of non-commuting observables.

We have already seen in problem set 4 that the Gaussian wave packet and the ground state of the simple harmonic oscillator are examples of minimum uncertainty states that saturate the bound, $\Delta x \Delta p = \frac{\hbar}{2}$.

> **Exercise 3.6.1:** Consider a physical observable which can take only two possible values and represented by operator $\hat{A}$, and a second physical observable which can, again, take only two possible values and represented by operator $\hat{B}$. The eigenstates $|a_1\rangle$ and $|a_2\rangle$ of $\hat{A}$ and $|b_1\rangle$ and $|b_2\rangle$ of $\hat{B}$ are related by $|a_1\rangle = \frac{3}{5}|b_1\rangle + \frac{4}{5}|b_2\rangle$ and $|a_2\rangle = \frac{4}{5}|b_1\rangle - \frac{3}{5}|b_2\rangle$.
>
> 1. A measurement of $\hat{A}$ (i.e., the observable represented by operator $\hat{A}$) produces a_1. What is the state of the system immediately after the measurement?
> 2. Right after the measurement of $\hat{A}$ in part (1), a following measurement of $\hat{B}$ is done. What are the possible outcomes and with what probability?
> 3. Right after the measurement of $\hat{B}$ in part (2), a following measurement of $\hat{A}$ is done. What is the probability for obtaining a_1?
> 4. Question (3) was answered without knowing the outcome in the measurement of $\hat{B}$ in part (2). Suppose b_2 was measured in part (2). What would have been the probability for measuring a_1 in part (3)?

3.7 Time evolution of expectation value of an operator and Ehrenfest's theorem

Consider an observable $\hat{O}$. How does its expectation value in state $|\psi(t)\rangle$ evolve in time? Starting with $\langle\hat{O}\rangle = \langle\psi(t)|\hat{O}|\psi(t)\rangle$, and using the Schrödinger equation $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$ and its adjoint $-i\hbar\frac{\partial}{\partial t}\langle\psi(t)| = \langle\psi(t)|\hat{H}$, we have

$$\begin{aligned}\frac{d}{dt}\langle\hat{O}\rangle &= \langle\psi(t)|\frac{\partial\hat{O}}{\partial t}|\psi(t)\rangle + \langle\psi(t)|\hat{O}\frac{\partial}{\partial t}|\psi(t)\rangle + \frac{\partial}{\partial t}(\langle\psi(t))|\hat{O}|\psi(t)\rangle \\ &= \langle\psi(t)|\frac{\partial\hat{O}}{\partial t}|\psi(t)\rangle - \frac{i}{\hbar}\langle\psi(t)|\hat{O}\hat{H}|\psi(t)\rangle + \frac{i}{\hbar}\langle\psi(t))|\hat{H}\hat{O}|\psi(t)\rangle\rangle\end{aligned}$$

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$$
\begin{aligned}
&= \langle\psi(t)|\frac{\partial\hat{O}}{\partial t}|\psi(t)\rangle - \frac{i}{\hbar}\langle\psi(t)|\hat{O}\hat{H}|\psi(t)\rangle + \frac{i}{\hbar}\langle\psi(t))|\hat{H}\hat{O}|\psi(t)\rangle \\
&= \langle\frac{\partial\hat{O}}{\partial t}\rangle + \frac{i}{\hbar}\langle[\hat{H},\ \hat{O}]\rangle. \qquad (3.7.1)
\end{aligned}
$$

Observe that the expectation value of an operator that does not explicitly depend on time and commutes with the Hamiltonian operator is constant in time. The expression given by (3.7.1) is called Ehrenfest's theorem.

> **Question 3.7.1:** Why is that the expectation value of an operator that does not explicitly depend on time and commutes with the Hamiltonian operator constant in time?

In particular, for $\hat{O} = \hat{\vec{p}}$, and a Hamiltonian that is time-independent,

$$
\begin{aligned}
\frac{d}{dt}\langle\hat{\vec{p}}\rangle &= \frac{i}{\hbar}\langle[\frac{\hat{\vec{p}}^2}{2m} + V(\hat{\vec{r}}),\ \hat{\vec{p}}]\rangle = \frac{i}{\hbar}\langle[V(\hat{\vec{r}}),\ -i\hbar\vec{\nabla}]\rangle = \frac{i}{\hbar}\langle i\hbar\vec{\nabla}V(\hat{\vec{r}})\rangle \\
&= -\langle\vec{\nabla}V(\hat{\vec{r}})\rangle. \qquad (3.7.2)
\end{aligned}
$$

Recalling that $\frac{d}{dt}\vec{p} = -\vec{\nabla}V(\vec{r})$ in classical mechanics, we see that the time evolution of the expectation value of momentum follows the classical equation with the expectation value of the negative gradient of the potential. Observe that $\frac{d}{dt}\vec{p} = -\vec{\nabla}V(\vec{r})$ is just Newton's second law, which governs classical mechanics, and, therefore, quantum mechanics contains all of classical mechanics.

3.8 The simple harmonic oscillator

Consider a particle of mass m in the one-dimensional simple harmonic oscillator. The Hamiltonian operator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \qquad (3.8.1)$$

Can we construct the wavefunctions without solving the Schrödinger equation? The answer is yes. What is the fundamental physics that could allow us to construct the wavefunctions without using the Schrödinger equation? It is the commutation relations between position and momentum. Indeed, one can do all of quantum mechanics starting with the commutation relations and without referring to the Schrödinger equation.

Our starting point now is that we are not going to be attempting to write the Hamiltonian in either the position or the momentum representations. We simply think of $\hat{H}$ as an operator

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that satisfies commutation relations (with position and momentum operators) that follow from the commutation relations of the position and momentum operators.

3.8.1 Raising and lowering operators

Suppose we trade $\hat{x}$ and $\hat{p}$ for two other operators $\hat{a}$ and $\hat{a}^\dagger$ such that

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right). \tag{3.8.2}$$

Note that $\hat{a}$ and $\hat{a}^\dagger$ are not Hermitian, since $\hat{a}^\dagger \neq \hat{a}$. The commutator between $\hat{a}$ and $\hat{a}^\dagger$ is $[\hat{a},\, \hat{a}^\dagger] = \frac{m\omega}{2\hbar}\frac{i}{m\omega}\left([\hat{p}, \hat{x}] - [\hat{x}, \hat{p}]\right) = \frac{i}{2\hbar}\left(-i\hbar - i\hbar\right]) = 1$. Therefore,

$$[\hat{a},\, \hat{a}^\dagger] = 1. \tag{3.8.3}$$

Next, we want to rewrite $\hat{H}$ in terms of $\hat{a}$. First, solving (3.8.2) for $\hat{x}$ and $\hat{p}$,

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^\dagger + \hat{a}\right), \qquad \hat{p} = i\sqrt{\frac{m\omega\hbar}{2}}\left(\hat{a}^\dagger - \hat{a}\right), \tag{3.8.4}$$

which in (3.8.1) gives

$$\begin{aligned}
\hat{H} &= \frac{1}{2m}\left(i\sqrt{\frac{m\omega\hbar}{2}}\left(\hat{a}^\dagger - \hat{a}\right)\right)^2 + \frac{1}{2}m\omega^2\left(\sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^\dagger + \hat{a}\right)\right)^2 \\
&= -\frac{1}{4}\hbar\omega\left(\hat{a}^{\dagger 2} + \hat{a}^2 - \hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger\right) + \frac{1}{4}\hbar\omega\left(\hat{a}^{\dagger 2} + \hat{a}^2 + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) \\
&= \frac{1}{2}\hbar\omega\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) = \frac{1}{2}\hbar\omega\left(2\hat{a}^\dagger\hat{a} + [\hat{a},\, \hat{a}^\dagger]\right) = \frac{1}{2}\hbar\omega\left(2\hat{a}^\dagger\hat{a} + 1\right) \\
&= \left(\hat{N} + \frac{1}{2}\right)\hbar\omega,
\end{aligned} \tag{3.8.5}$$

where we have defined

$$\hat{N} = \hat{a}^\dagger\hat{a}. \tag{3.8.6}$$

Now $\hat{N}$ is Hermitian and, therefore, it has orthonormal eigenstates with corresponding eigenvalues. If we solve the eigenvalue problem for the operator $\hat{N}$, we will also have solved the eigenvalue problem for the Hamiltonian operator $\hat{H}$. Therefore, we seek the eigenstate $|n\rangle$ and the corresponding eigenvalue n that solves

$$\hat{N}|n\rangle = n|n\rangle. \tag{3.8.7}$$

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Consider the commutator between $\hat{N}$, $\hat{a}$, and $\hat{a}^\dagger$,

$$[\hat{N},\,\hat{a}] = [\hat{a}^\dagger\hat{a},\,\hat{a}] = \hat{a}^\dagger[\hat{a},\,\hat{a}] + [\hat{a}^\dagger,\,\hat{a}]\hat{a} = -\hat{a} \tag{3.8.8}$$

and

$$[\hat{N},\,\hat{a}^\dagger] = [\hat{a}^\dagger\hat{a},\,\hat{a}^\dagger] = \hat{a}^\dagger[\hat{a},\,\hat{a}^\dagger] + [\hat{a}^\dagger,\,\hat{a}^\dagger]\hat{a} = \hat{a}^\dagger. \tag{3.8.9}$$

Question 3.8.1: $\hat{a}|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue (A). n, (B). $n+1$, (C). $n-1$, (D). $-n$, (D). 0

Now, given $\hat{N}|n\rangle = n|n\rangle$ and $[\hat{N},\,\hat{a}] = -\hat{a}$, we have

$$\hat{N}(\hat{a}|n\rangle) = \left([\hat{N},\,\hat{a}] + \hat{a}\hat{N}\right)|n\rangle = \left(-\hat{a} + \hat{a}\hat{N}\right)|n\rangle = (-\hat{a} + \hat{a}n)\,|n\rangle = (n-1)\,(\hat{a}|n\rangle)\,. \tag{3.8.10}$$

Therefore, $\hat{a}|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $n-1$. Because $\hat{a}$ applied on $|n\rangle$ lowers the eigenvalue of $\hat{N}$ by one unit, we call $\hat{a}$ a lowering operator. Moreover,

$$\hat{N}\left(\hat{a}^\dagger|n\rangle\right) = \left([\hat{N},\,\hat{a}^\dagger] + \hat{a}^\dagger\hat{N}\right)|n\rangle = \left(\hat{a}^\dagger + \hat{a}^\dagger\hat{N}\right)|n\rangle = \left(\hat{a}^\dagger + \hat{a}^\dagger n\right)|n\rangle = (n+1)\left(\hat{a}^\dagger|n\rangle\right). \tag{3.8.11}$$

Therefore, $\hat{a}^\dagger|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $n+1$, and we call $\hat{a}^\dagger$ a raising operator.

3.8.2 Normalized number state vectors

Because the energy levels of the simple harmonic oscillator in one dimension are not degenerate, the states $|n-1\rangle$ and $\hat{a}|n\rangle$ must be physically the same, and could only differ by a constant factor. Thus, we write

$$|n-1\rangle = c_n\hat{a}|n\rangle, \tag{3.8.12}$$

where c_n is a constant. The constant c_n follows from the normalization of the states. In particular, considering the matrix element of $\hat{N}$,

$$\langle n|\hat{N}|n\rangle = \langle n|\hat{a}^\dagger\hat{a}|n\rangle = |\hat{a}|n\rangle|^2\,. \tag{3.8.13}$$

On the other hand,

$$\langle n|\hat{N}|n\rangle = n\langle n|n\rangle \tag{3.8.14}$$

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for normalized $|n\rangle$. Therefore,

$$1 = \langle n-1|n-1\rangle = |c_n|^2\langle n|\hat{N}|n\rangle = |c_n|^2 n, \tag{3.8.15}$$

or, choosing real c_n, we have $c_n = \frac{1}{\sqrt{n}}$, and

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle. \tag{3.8.16}$$

Similarly,

$$|n+1\rangle = d_n\hat{a}^\dagger|n\rangle, \tag{3.8.17}$$

where d_n is a constant. But

$$1 = \langle n+1|n+1\rangle = |d_n|^2\langle n|\hat{a}\hat{a}^\dagger|n\rangle = |d_n|^2\langle n|[\hat{a},\,\hat{a}^\dagger] + \hat{a}^\dagger\hat{a}|n\rangle = |d_n|^2(1+n). \tag{3.8.18}$$

Therefore, choosing real d_n, we have $d_n = \frac{1}{\sqrt{n+1}}$, and

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \tag{3.8.19}$$

Because the eigenvalue n of $\hat{N}$ satisfies

$$n = \langle n|\hat{N}|n\rangle = |\hat{a}|n\rangle|^2 \geq 0, \tag{3.8.20}$$

it is positive definite or zero. Because $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and we cannot have negative n, we require $\hat{a}|0\rangle = 0$. Therefore, the ground state is $|0\rangle$. The excited states are then constructed by applying the raising operator on $|0\rangle$. In particular,

$$\hat{a}^\dagger|0\rangle = \sqrt{1}|1\rangle \quad \text{or} \quad |1\rangle = \hat{a}^\dagger|0\rangle, \tag{3.8.21}$$

$$\hat{a}^\dagger|1\rangle = \sqrt{2}|2\rangle \quad \text{or} \quad |2\rangle = \frac{1}{\sqrt{2}}\hat{a}^\dagger|1\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2}}|0\rangle, \tag{3.8.22}$$

$$\hat{a}^\dagger|2\rangle = \sqrt{3}|3\rangle \quad \text{or} \quad |3\rangle = \frac{1}{\sqrt{3}}\hat{a}^\dagger|2\rangle = \frac{(\hat{a}^\dagger)^3}{\sqrt{3!}}|0\rangle, \tag{3.8.23}$$

and continuing,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \ldots \tag{3.8.24}$$

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The matrix elements of position and momentum follow from (3.8.4), (3.8.16), and (3.8.19),

$$\langle n'|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle n'|\left(\hat{a}^\dagger + \hat{a}\right)|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{n',n+1} + \sqrt{n}\,\delta_{n',n-1}\right) \tag{3.8.25}$$

and

$$\langle n'|\hat{p}|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\langle n'|\left(\hat{a}^\dagger - \hat{a}\right)|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\left(\sqrt{n+1}\,\delta_{n',n+1} - \sqrt{n}\,\delta_{n',n-1}\right). \tag{3.8.26}$$

Note that both $\hat{x}$ and $\hat{p}$ have only off-diagonal matrix elements, which is consistent with $\hat{H}$ or $\hat{N}$ having diagonal elements in the basis with state vectors $|n\rangle$, since $\hat{H}$ does not commute with either one of $\hat{x}$ or $\hat{p}$.

Question 3.8.2: Why should the ground state be $|0\rangle$ with eigenvalue $n = 0$ for $\hat{N}$? Why not a nonzero constant?

Question 3.8.3: Consider a set of operators $\hat{A}$, $\hat{R}$, and $\hat{L}$ such that $\hat{A}|a\rangle = a|a\rangle$, $[\hat{A}, \hat{R}] = r\hat{R}$, and $[\hat{A}, \hat{L}] = -l\hat{L}$ with $l, r > 0$. Which of the following statements are valid. (A). $|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue a. (B). $\hat{L}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $a - l$. (C). $\hat{R}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $a + r$. (D). $\hat{L}$ is a lowering operator. (E). $\hat{R}$ is a raising operator.

3.8.3 Wavefunctions in position representation

First, the eigenvalues follow from $\hat{H}|n\rangle = \left(\hat{N} + \frac{1}{2}\right)\hbar\omega = \left(n + \frac{1}{2}\right)\hbar\omega$. Therefore,

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \ldots \tag{3.8.27}$$

Next, we want to construct the wavefuctions corresponding to these energy levels in position representation. In particular, the stationary wavefunctions of definite energy are

$$\psi_n(x) = \langle x|n\rangle. \tag{3.8.28}$$

First, let us construct the ground state wavefunction. Using $\hat{a}|0\rangle = 0$,

$$0 \;=\; \hat{a}|0\rangle = \langle x|\hat{a}|0\rangle = \sqrt{\frac{m\omega}{2\hbar}}\langle x\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)|0\rangle$$

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$$= \sqrt{\frac{m\omega}{2\hbar}}\left(x\langle x|0\rangle + \frac{i}{m\omega}(-i\hbar\frac{\partial}{\partial x})\langle x|0\rangle\right). \tag{3.8.29}$$

Thus, we have a differential equation for $\langle x|0\rangle$,

$$\frac{\hbar}{m\omega}\frac{\partial}{\partial x}\langle x|0\rangle + x\langle x|0\rangle = 0. \tag{3.8.30}$$

The solution is

$$\langle x|0\rangle = \mathcal{A}e^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.31}$$

Normalizing,

$$\langle x|0\rangle = \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.32}$$

The wavefunctions for the excited states are then obtained by applying the raising operator. First let us determine the matrix element of $\langle x'|\hat{a}^\dagger|x''\rangle$,

$$\begin{aligned}
\langle x'|\hat{a}^\dagger|x''\rangle &= \langle x'|\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)|x''\rangle = \sqrt{\frac{m\omega}{2\hbar}}\left(\langle x'|\hat{x}|x''\rangle - \frac{i}{m\omega}\langle x'|\hat{p}|x''\rangle\right) \\
&= \sqrt{\frac{m\omega}{2\hbar}}\left(x''\delta(x'-x'') - \frac{i}{m\omega}\left(-i\hbar\frac{\partial\delta(x'-x'')}{\partial x'}\right)\right) \\
&= \frac{1}{\sqrt{2}\sigma}\left(x''\delta(x'-x'') - \sigma^2\frac{\partial\delta(x'-x'')}{\partial x'}\right) \\
&= \frac{1}{\sqrt{2}\sigma}\left(x''\delta(x'-x'') + \sigma^2\frac{\partial\delta(x'-x'')}{\partial x''}\right) \\
&= \delta(x'-x'')\frac{1}{\sqrt{2}\sigma}\left(x'' - \sigma^2\frac{\partial}{\partial x''}\right),
\end{aligned} \tag{3.8.33}$$

where we have defined

$$\sigma \equiv \sqrt{\frac{\hbar}{m\omega}} \tag{3.8.34}$$

for simplicity of notation. Inserting the above expression for the matrix elements and integrating,

$$\begin{aligned}
\langle x|n\rangle &= \langle x|\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle = \frac{1}{\sqrt{n!}}\int dx_1 dx_2 ... dx_{n-1}\langle x|\hat{a}^\dagger|x_1\rangle\langle x_1|\hat{a}^\dagger|x_2\rangle\langle x_2|\cdots|x_{n-1}\rangle\langle x_{n-1}|\hat{a}^\dagger|x_n\rangle\langle x_n|0\rangle \\
&= \frac{1}{\sqrt{n!}}\left(\frac{1}{\sqrt{2}\sigma}\left(x - \sigma^2\frac{\partial}{\partial x}\right)\right)^n\langle x|0\rangle \\
&= \frac{1}{\sqrt{n!2^n}\sigma^n}\left(x - \sigma^2\frac{\partial}{\partial x}\right)^n\left(\frac{1}{\pi\sigma^2}\right)^{1/4} e^{-\frac{1}{2\sigma^2}x^2}
\end{aligned} \tag{3.8.35}$$

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$$= \frac{1}{\sqrt{\sqrt{\pi} n! 2^n \, \sigma^{n+\frac{1}{2}}}} \left(x - \sigma^2 \frac{\partial}{\partial x} \right)^n e^{-\frac{1}{2\sigma^2} x^2}. \tag{3.8.36}$$

If we define

$$\xi = \frac{x}{\sigma}, \tag{3.8.37}$$

then

$$\langle x|n\rangle = \frac{1}{\sqrt{\sqrt{\pi} n! 2^n \sigma}} \left(\xi - \frac{\partial}{\partial \xi} \right)^n e^{-\frac{1}{2}\xi^2} \tag{3.8.38}$$

This reproduces the wavefunctions obtained after solving the Schrodinger equation. In particular, for the ground and first excited states,

$$\langle x|0\rangle = \frac{1}{\sqrt{\sqrt{\pi}\sigma}} e^{-\frac{1}{2}\xi^2} = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} x^2} \tag{3.8.39}$$

and

$$\langle x|1\rangle = \frac{1}{\sqrt{\sqrt{\pi} 2\sigma}} \left(\xi - \frac{\partial}{\partial \xi} \right) e^{-\frac{1}{2}\xi^2} = \frac{1}{\sqrt{\sqrt{\pi} 2\sigma}} 2\xi e^{-\frac{1}{2}\xi^2} = \sqrt{2} \left(\frac{m^3\omega^3}{\pi\hbar^3} \right)^{1/4} x e^{-\frac{m\omega}{2\hbar} x^2}. \tag{3.8.40}$$

3.8.4 Classical simple harmonic oscillator

The probability density for finding the particle at point x in a stationary states $\langle x|n\rangle$ does not evolve in time. In other words, it does not oscillate. Can we construct a quantum state that exhibits a classical-like oscillation?

First let us briefly overview the classical equations of motion of the simple harmonic oscillator in terms of variables that would resemble the quantum oscillator. The Hamiltonian (or total energy) for a classical simple harmonic is

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2, \tag{3.8.41}$$

which is the same as (3.8.1) before the position and momentum are promoted to operators. The

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time evolution of x and p are given by Hamilton's equations,

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m} \tag{3.8.42}$$

and

$$\dot{p} = -\frac{\partial H}{\partial x} = -m\omega^2 x. \tag{3.8.43}$$

Let us trade x and p for a new complex variable α defined by

$$\sqrt{\hbar\omega}\,\alpha = \sqrt{\frac{m\omega^2}{2}}x + \frac{i}{\sqrt{2m}}p. \tag{3.8.44}$$

With this, we rewrite the Hamiltonian,

$$H = \hbar\omega|\alpha|^2. \tag{3.8.45}$$

Moreover,

$$x = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R, \qquad p = \sqrt{2m\hbar\omega}\,\alpha_I \tag{3.8.46}$$

where α_R and α_I are the real and imaginary parts of α.

The time evolution of α is then

$$\begin{aligned}
\sqrt{\hbar\omega}\,\dot{\alpha} &= \sqrt{\frac{m\omega^2}{2}}\dot{x} + \frac{i}{2m}\dot{p} = \sqrt{\frac{m\omega^2}{2}}\frac{p}{m} + \frac{i}{\sqrt{2m}}(-m\omega^2 x) \\
&= \sqrt{\frac{m\omega^2}{2}}\frac{1}{m}\sqrt{2m\hbar\omega}\,\alpha_I - \frac{i}{\sqrt{2m}}m\omega^2\sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R \\
&= -i\sqrt{\hbar\omega^3}\,(\alpha_R + i\alpha_I) = -i\sqrt{\hbar\omega^3}\,\alpha.
\end{aligned} \tag{3.8.47}$$

or

$$\dot{\alpha} = -i\omega\alpha. \tag{3.8.48}$$

The solution is

$$\alpha = \alpha_0\, e^{-i\omega t}. \tag{3.8.49}$$

As a specific example for a real α_0, then $\alpha_R = \alpha_0 \cos\omega t$ and $\alpha_I = \alpha_0 \sin\omega t$. In this case, $x = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_0 \cos\omega t$ and $p = \sqrt{2m\hbar\omega}\,\alpha_I = \sqrt{2m\hbar\omega}\,\alpha_0 \sin\omega t$. Therefore, the real and imaginary parts of α are related to the position and momentum of the oscillator. In particular,

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expressing the Hamiltonian of the classical oscillator in terms of α,

$$H = |\alpha|^2 \hbar\omega. \tag{3.8.50}$$

3.8.5 The quantum simple harmonic oscillator and coherent state

Question 3.8.4: What is the frequency of oscillation of the probability density for a simple harmonic oscillator in state $|n\rangle$? (A). ω, (B). $\frac{E_n}{\hbar}$, (C). $\frac{E_n - E_{n-1}}{\hbar}$, (D). 0.

Question 3.8.5: What could we do to construct a state whose time evolution could look like a classical oscillator?

Returning back to the quantum oscillator, and noting the expression $H = |\alpha|^2\hbar\omega$ for the classical oscillator, we observe the expectation value of $\hat{H}$,

$$\langle \hat{H} \rangle = \left(\langle \hat{a}^\dagger \hat{a} \rangle + \frac{1}{2} \right) \hbar\omega \tag{3.8.51}$$

has a similar form as the classical H if we identify $\langle \hat{a}^\dagger \hat{a} \rangle$ with $|\alpha|^2$ and redefine the classical potential energy such that $H \to H = \left(|\alpha|^2 + \frac{1}{2} \right) \hbar\omega$. Thus, we seek a quantum state $|\alpha\rangle$ which gives $\langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = |\alpha|^2$. This is possible if we could construct a state $|\alpha\rangle$ such that

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle. \tag{3.8.52}$$

The state $|\alpha\rangle$ that satisfies this equation is called coherent state (or Glauber state, after the recently late Roy Glauber of Harvard). It is a quantum state whose dynamics resembles that of a classical oscillator. We will investigate it in problem set 6.

Question 3.8.6: What is the frequency of oscillation of the probability density of a coherent state of energy $\frac{1}{2}\hbar\omega$? What about for a coherent state of energy $E > \frac{1}{2}\hbar\omega$?

Exercise 3.8.1: Using the raising and lowering operators, answer the following questions for a particle of mass m in the one-dimensional harmonic oscillator with potential $V(x) = \frac{1}{2} m\omega^2 x^2$.

1. The uncertainty in position, Δx, in stationary state $|n\rangle$.
2. The uncertainty in momentum, Δp, in stationary state $|n\rangle$.
3. Check whether the product $\Delta x \, \Delta p$ obeys the uncertainty principle.

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4. Rewrite the Hamiltonian $\hat{H}$ in terms of $\hat{\alpha}_R$ and $\hat{\alpha}_I$ defined by $\hat{x} = \sqrt{\frac{2\hbar}{m\omega}}\,\hat{\alpha}_R$ and $\hat{p} = \sqrt{2m\hbar\omega}\,\hat{\alpha}_I$. Rewrite the uncertainty relation in stationary state $|n\rangle$ obtained in part (3) for $\Delta\alpha_R\Delta\alpha_I$.

5. When the oscillator is not in a stationary state, or an eigenstate of the Hamiltonian operator, the uncertainty would take other values than those for individual stationary states obtained above. Show the boundary that separates the region that satisfies the uncertainty principle and the region that does not on a plot of $\Delta\alpha_I$ versus $\Delta\alpha_R$. Indicate the points $(\Delta\alpha_R,\ \Delta\alpha_I)$ on the plane for the first four stationary states corresponding to $n = 0, 1, 2,$ and 3.

Exercise 3.8.2: A coherent state (or Glauber state) $|\alpha\rangle$ of the simple harmonic oscillator is an eigenstate of the lowering operator, $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$.

1. Determine the coherent state as a superposition of the stationary states $|n\rangle$ and normalize it.

2. What is the probability that a system in a coherent state would be found in stationary state $|n\rangle$? What kind of probability distribution function of n is it?

3. Determine the uncertainties Δx, $\Delta\alpha_R$, Δp, and $\Delta\alpha_R$, using the raising and lowering operators, in the coherent state.

4. Show the point for uncertainties $(\Delta\alpha_R,\ \Delta\alpha_I)$ for the coherent state on the plot you made in part (5). Comparing $\Delta\alpha_R$, $\Delta\alpha_I$, and their product, what is special about the coherent state?

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